# Competency Based Learning Material


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|---|---|
| Sector: **AGRICULTURE, FORESTRY AND FISHERY** | |
| Qualification : **ORGANIC AGRICULTURE PRODUCTION NC II** | |
| Unit of Competency: **PRODUCE ORGANIC CONCOCTION AND EXTRACT** | |
| Module Title **PRODUCING ORGANIC CONCOCTION AND EXTRACT** | |
| **MDM SAGAY COLLEGE INC.** | |

# ORGANIC AGRICULTURE PRODUCTION NC II

The **ORGANIC AGRICULTURE PRODUCTION NC II** Qualification consists of competencies that a person must achieve to produce organic farm products such as chicken and vegetables including producing of organic supplements such as fertilizer, concoctions and extracts. It has two (2) elective competencies which are on raising organic hogs and raising organic small ruminants.

This Qualification is packaged from the competency map of the Agri-Fishery Sector.

The units of competency comprising this qualification includes the following:

| CODE | BASIC COMPETENCIES |
|---|---|
| 500311105 | Participate in workplace communication |
| 500311106 | Work in a team environment |
| 500311107 | Practice career professionalism |
| 500311108 | Practice occupational health and safety procedures |

| CODE | COMMON COMPETENCIES |
|---|---|
| AFF 321201 | Apply safety measures in farm operations |
| AFF 321202 | Use farm tools and equipment |
| AFF 321203 | Perform estimation and calculations |
| TRS311201 | Develop and update industry knowledge |
| AGR321205 | Perform record keeping |

| CODE | CORE COMPETENCIES |
|---|---|
| AGR612301 | Raise organic chicken |
| AGR611306 | Produce organic vegetables |
| AGR611301 | Produce organic fertilizer |
| AGR611302 | Produce organic concoctions and extract |
| **CODE** | **ELECTIVE COMPETENCIES** |
| AGR612302 | Raise organic hogs |
| AGR612303 | Raise organic small ruminants |

# HOW TO USE THIS MODULE

Welcome to the Module on Producing Organic Concoction and extract. This module contains training materials and activities for you to complete.

The unit of competency on *Produce organic concoction and extract* contains knowledge, skills and attitudes required for an Organic Agriculture Production NC II course.

You are required to go through a series of learning activities in order to complete each of the learning outcomes of the module. In each learning outcome there are **Information Sheets, Operation Sheets**, **Job Sheets** and **Task Sheets**. Follow these activities on your own and answer the Self-Check at the end of each learning activity.

If you have questions, don't hesitate to ask your trainer for assistance.

**Recognition of Prior Learning (RPL)**

You may already have some of the knowledge and skills covered in this module because you have:

- been working for some time
- already have completed training in this area.

If you can demonstrate to your trainer/instructor that you are competent in a particular skill or skills, talk to him/her about having them formally recognized so you don't have to do the same training again. If you have a qualification or Certificate of Competency from previous trainings show it to your trainer. If the skills you acquired are still current and relevant to this module, they may become part of the evidence you can present for RPL. If you are not sure about the currency of your skills, discuss it with your trainer.

After completing this module ask your trainer to assess your competency. Result of your assessment will be recorded in your competency profile. All the learning activities are designed for you to complete at your own pace.

Inside this module you will find the activities for you to complete followed by relevant information sheets for each learning outcome. Each learning outcome may have more than one learning activity.

## MODULE CONTENT

Program/Course : Organic Agriculture Production NC II
Unit of Competency : Produce Organic Concoction and Extract
Module : Producing Organic Concoction and Extract

**INTRODUCTION:**

This module contains information and suggested learning activities on Producing Organic Concoction and Extract. It includes activities and materials on Organic Agriculture Production NC II

Completion of this module will help you better understand the succeeding module on the producing organic concoction and extract.

This module consists of 3 learning outcomes. Each learning outcome contains learning activities supported by each instruction sheets. Before you perform the instructions, read the information sheets and answer the self-check and activities provided to ascertain to yourself and your trainer that you have acquired the knowledge necessary to perform the skill portion of the particular learning outcome.

Upon completion of this module, report to your trainer for assessment to check your achievement of knowledge and skills requirement of this module. If you pass the assessment, you will be given a certificate of completion.

## SUMMARY OF LEARNING OUTCOMES:

Upon completion of the module you should be able to:

LO1 Prepare for the production of various concoctions
LO2 Process concoctions
LO3 Package concoctions

**Learning Outcome #1 Prepare for the production of various concoctions**

**Assessment Criteria:**

Work and storage areas are cleaned, sanitized and secured.

Raw materials used are cleaned and freed from synthetic chemicals

Tools, materials and equipment used are cleaned, freed from contaminations and must be of "food grade" quality

Personal hygiene are observed according to OHS procedures.

**Contents:**
Storage Areas
Raw materials
Tools, materials and equipment
Personal hygiene

**Tools, Materials and Equipment and Facilities**

kangkong, camote tops, alugbati, malunggay, banana trunks, bamboo shoots and other fast growing green plants, Molasses/mascuvado/ brown sugar Ripe and sweet fruits but not limited to banana, papaya, watermelon, ampalaya , tomato Trash Fish and gills, scales, offal of big fishes, golden kuhol meat Garlic, ginger, Pure coconut vinegar animal bones, egg shell, sea shell, kuhol shell 1 kl. Cooked, cool rice900 ml. fresh milk, 100 ml clear liquid from fermented rice, Plastic pail, Wooden ladle, Manila paper or cheese cloth, String or rubber bands, Weighing scale, Chopping board, Knife Marker, Strainer or nylon screen,

**References:**


https://www.youtube.com/watch?v=GouWt_DM544

https://www.youtube.com/watch?v=Qr3gZAbnrHM

https://www.youtube.com/watch?v=W58mZaJCArA

https://www.youtube.com/watch?v=peJeyMV2Plc https://www.youtube.com/watch?v=lspuVsEW3vY

https://www.youtube.com/watch?v=6a1YezkoLjs

https://www.youtube.com/watch?v=A40KLz6fRk8

https://www.youtube.com/watch?v=UXysXSFuME0

https://www.youtube.com/watch?v=vCFkmuUL-S0

https://www.youtube.com/watch?v=XzdeuMz3MZ4
http://organic.da.gov.ph/images/IECs/FPJ.pdf

## LEARNING EXPERIENCES/ACTIVITIES

## LO # 1 Prepare for the production of various concoctions

| Learning Activities | Special Instructions |
|---|---|
| Read the attached Information Sheet 4.1-1 on storage area | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 4.1-1 on storage area | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 4.1-1 on storage area | |
| Perform Job Sheet 4.1-1 on storage area | Observe the demonstration of your trainer and check the procedures while performing the operation. |
| Evaluate performance using performance checklist on 4.1-1 storage area | |
| Read the attached Information Sheet 4.1-2 on Raw materials | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 4.1-2 on Raw materials | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 4.1-2 on Raw materials | |
| Read the attached Information Sheet 4.1-3 on Tools, Materials and Equipment | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 4.1-3 on Tools, Materials and Equipment | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 4.1-3 on Tools, Materials and Equipment | |

| | |
|---|---|
| Perform Job Sheet 4.1-3 on Tools, Materials and Equipment | Observe the demonstration of your trainer and check the procedures while performing the operation. |
| Evaluate performance using performance checklist on 4.1-3 on Tools, Materials and Equipment | |
| Read the attached Information Sheet 4.1-4 on Personal Hygiene | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 4.1-4 on Personal Hygiene | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 4.1-4 on Personal Hygiene | |
| Perform Job Sheet 4.1-4 on Personal Hygiene | Observe the demonstration of your trainer and check the procedures while performing the operation. |
| Evaluate performance using performance checklist on 4.1-4 on Personal Hygiene | |

***After doing this activity, you are now ready to proceed to the next module on Process concoctions.***

# INFORMATION SHEET 4.1-1
# Storage Area

**Learning Objective:** After reading this information sheet, you should be able to secure the Storage Areas

**Introduction:**

Secure the building and storage site, only you and your authorized employees should have access to the storage area. Keep the storage unit locked at all times, except when it is under the direct supervision of a person authorized for entry. For extra security, install a fence around the storage area and lock the gate. Consider installing security lighting and an alarm system.

**Basic Safety Guidelines**:

- Never let anyone eat, drink, or smoke in the storage facility.
- Store pesticides in their original labeled containers. Never store pesticides in milk jugs, soft drink bottles, fruit jars, or medicine bottles.
- Do not store pesticides with or near food, medicine, cleaning supplies, fertilizers, seed, or animal feed.
- Do not keep gasoline, kerosene, or other combustible materials with pesticides.
- Make sure pesticides are not kept near operations that present a fire hazard, such as burning and welding.
- Do not leave any pesticide container in full sun or next to a heater.
- Store pesticides on metal shelves with a lip or on wood shelves covered with plastic or chemically-resistant epoxy paint. Leak-proof plastic trays on shelves work well. Do not store pesticides on the floor. Use pallets under large containers/bags.
- Keep the storage area neat and clean at all times. Keep the area free of debris such as waste paper, rags, or used cardboard boxes, which may provide an ignition source. Clean up any spills immediately.
- Store dry formulations on the highest shelves. Store liquids and glass containers on the lowest level. This will prevent contamination in case a liquid container leaks.

**Warning/Emergency response signs**

Place signs indicating “Danger Pesticides – Keep Out – No Smoking” at all storage entries. Consider posting signs in Spanish as well as English. Some state laws require additional signage indicating who is responsible for the pesticide storage and who to call for emergencies. This type of sign should have at least two emergency phone numbers – the owner should not be the sole contact in an accident. The National Fire Protection Association (NFPA) 704 standard provides a way to communicate the potential hazards of storing hazardous chemicals through the posting of a diamond shape or square-on-point shape sign. The sign addresses the health, flammability, instability, and related hazards associated with short-term exposures that are most likely to occur as a result of fire, spill, or similar emergency. The 704 standard is applicable to industrial, commercial, and institutional facilities that manufacture, handle, or store hazardous materials. For more details on this standard, refer to the National Fire Protection Association web

## SELF CHECK 4.1-1

## Storage Area

**True or False**

1. Let anyone eat, drink, or smoke in the storage facility.
2. Never store pesticides in milk jugs, soft drink bottles, fruit jars, or medicine bottles.
3. Do not leave any pesticide container in full sun or next to a heater.
4. Keep the storage area neat and clean at all times.
5. Do not keep the area free of debris such as waste paper, rags, or used cardboard boxes, which may provide an ignition source.

# ANSWER KEY 4.1-1
# Storage Area

## True or False

1. False
2. True
3. True
4. True
5. False

# JOB SHEET 4.1-1

## Storage Area

| |
|---|
| **Title: Securing the Storage Area** |
| **Performance Objective:** At the end of this module the student should be able to secure the storage area. |
| **Supplies/Materials:** PPE, Storage Facility |
| **Steps/Procedures:**<br><br>Wear the PPE<br><br>Install fence around the storage area<br><br>Install security lighting and an alarm system.<br><br>Keep pesticide container properly<br><br>Store pesticides on metal shelves with a lip or on wood shelves covered with plastic<br><br>Keep the storage area neat and clean at all times<br><br>Do housekeeping |
| **Assessment Method:**<br><br>Demonstration |

## PERFORMANCE CHECKLIST 4.1-1

### Storage Area

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee.. | | |
| 1. Wear the PPE? | | |
| 2. Install fence around the storage area? | | |
| 3. Install security lighting and an alarm system? | | |
| 4. Keep pesticide containers properly? | | |
| 5. Store pesticides on metal shelves with a lip or on wood shelves covered with plastic? | | |
| 6. Keep the storage area neat and clean at all times? | | |
| 7. Do housekeeping? | | |

## INFORMATION SHEET 4.1-2

## Raw materials

**Learning Objective:** After reading this information sheet, you should be able to determine the clean raw materials free from chemicals

**Introduction:**

All raw materials are cleaned before processing. The purpose is obviously to remove contaminants, which range from innocuous to dangerous. It is important to note that removal of contaminants is essential for protection of process equipment as well as the final consumer.

Choosing the raw materials for FPJ

You choose materials that are:

► Young and fresh

► Free from insect pests and diseases

► Abundant in the production area

► Free from chemical containments

**Materials needed in making FPJ**

■ Local plants that are fast growing like kangkong, legumes and grasses.

■ You can also use bamboo shoots, asparagus shoots, actively growing plant parts and young fruits of cucumber, squash, melon, watermelon, ampalaya and other cucurbits.

■ Weed species that are found growing in the production area, young leaves of trees, banana trunks, young leaves and fruits of stress tolerant crops are also good materials for FPJ.

■ You can use either crude sugar or molasses or whichever is available and can be bought at a lower price.

■ You will need basin, ceramic pot or plastic pail, net bag or cloth bag, paper or cloth for cover, string, stone as weight, bolo, chopping board, marking pen and glass jars.

**Steps in preparing raw materials for concoction.**

1. Collect the plant materials early in the morning while they are fresh and the microorganisms are still present. Do not wash the plant materials.

2. Cut the plant materials into small pieces so that the juice can be easily extracted.

3. Put 3 kg chopped plant materials in a basin, add 1 kg crude sugar or molasses, then mix thoroughly with your hands. Make sure that all plant materials are mixed with sugar so that the juice can be extracted easily.

4. Put the mixture in a net bag or cloth bag. This is done so that the extracted juice will ooze from all sides of the bag.

5. Put the bagged mixture in a ceramic pot or plastic pail, and put weight to compress the mixture. Stone is a good material used to weigh down the mixture.

6. Cover the pot or pail with paper or cloth, and secure with a string or rubber band. Paper or cloth is used as cover to allow some air to get inside the pot and for the gas that is being produced during the fermentation process to escape. On the cover, write the date of processing and the expected date of harvest.

7. Store the container with the bagged mixture in a cool dry shady place. Make sure that the storage area is not infested with cockroaches or mice, because they might feed on the mixture and contaminate the extract. In 7 days, plant juice is extracted and fermented. The plant extract will change its color from green to yellow, then to brown and will smell sweet and alcoholic.

8. After 7 days, lift the bagged mixture and squeeze hard to get the remaining extracts.

9. Collect the fermented extracts and preserve in dark colored glass jar. To cover the jar, use paper or cloth to allow the gas to escape during further fermentation, then, store in a cool, shady place. You may add the plant residue to the compost pile to hasten decomposition or you can apply it to the garden plots as source of organic matter. Use your FPJ more effectively if it is stored for another one week after completion

# SELF CHECK 4.2-1

## Raw Materials

**TRUE OR FALSE:**

1. Do not wash the plant materials.
2. Collect the plant materials early in the morning while they are fresh and the microorganisms are still present.
3. Make sure that all plant materials are mixed with sugar so that the juice can be extracted easily.

4. Put the bagged mixture in a ceramic pot or plastic pail, and put weight to compress the mixture.

5. Store the container with the bagged mixture in a cool dry shady place.

# ANSWER KEY 4.1-2

## Raw Materials

1. TRUE
2. TRUE
3. TRUE
4. TRUE
5. TRUE

## INFORMATION SHEET 4.1-3
## Tools, Materials and Equipment

**Learning Objective:** After reading this information sheet, you should be able to identify the tools, materials and equipment.

**Introduction:**

**PERSONAL PROTECTIVE EQUIPMENT** (PPE) Apron - made of cloth or plastic used as protection from any liquid materials and dirt when doing the fermentation procedure.

**Hair Net** - made of nets to cover the head to eliminate contamination of the fermented concoction.

**Gloves** - it is made up of rubber or plastic used to cover the hand to minimize the contamination of the fermented concoction.

**Face Mask** - made up of cloth or thin plastic used to cover the mouth eliminate contamination of the fermented concoction.

**Rubber Boots** - it is used to protect the feet from any clutter around the work place.

**MATERIALS**

**Fermented Plant Juice (FPJ)**
**Alugbati** - a plant that bears fruit that ranges in color from dark green to red.

**Bamboo Shoots** - new cane culms that come out of the ground - also known as bamboo sprouts

**Banana Trunks** - an inner part of the banana plant stems edible, healthy and rich in fibers.

**Kamote Tops** - a good source of protein, minerals, dietary fiber and nutrients such as calcium, magnesium, sodium, phosphorous, sulfur, iron and zinc.

**Kangkong** - a plant known in English as water spinach, river spinach, water morning glory and water convolvulus that gives a lot of health benefits.

**Molasses** - a dark color liquid and about two thirds as sweet as regular sugar used in fermentation process.

**Mascuvado** - a natural sugar from sugar cane and alternative raw material in fermentation process.

**Brown sugar** - a natural material also from sugar cane and alternative raw material in fermentation process.

**Fermented Fruit Juice (FFJ) at least three kinds of ripe fruits Banana** - a fruit that rich in potassium that provides instant energy.

**Papaya -** A contains enzyme called papain that aids digestion, high in fiber and water content, which help to prevent constipation and promote regularity and a healthy digestive tract. Watermelon - a delicious and refreshing fruit that contain 46 calories per cup, high in vitamin C and vitamin A.

**Ampalaya** - known as bitter melon for diabetes, stomach and intestinal problems, promote menstruation, and many other conditions

**Tomato** - a fruit that contains a chemical called lycopene, which is believed to play a role in preventing cancer.

**Fish Amino Acid (FAA) Golden Kuhol Meat** - known as Golden Apple Snail that highly invasive and cause damage to rice crops wihc can be a source of nitrogen fertilizer for plants and protein supplement to animals.

**Scales of all of big fishes** - these are the outer covering of fish which can be used in producing Amino Acid for plants and animals.

**Trash fish -** these are marine fish having little or no market value as human food but used sometimes in the production of fish meal simply waste of fish like head, bone, intestines, scales, gills, etc

**Oriental Herbal Nutrient (OHN) Chili** - a small hot-tasting pod of a variety of capsicum, used chopped (and often dried) in sauces, relishes, and spice powders.

**Garlic** - a strong-smelling pungent-tasting bulb, used as a flavoring in cooking and in herbal medicine.

**Ginger** - a hot fragrant spice made from the rhizome of a plant. It is chopped or powdered for cooking, preserved in syrup, or candied.

**Makabuhay** - a primary ingredient used to concoct preparations that would prevent spread of malaria, and may be used as cleanser for skin ulcer and skin wounds. Pure coconut vinegar - produced from the sap and water of coconut trees that is fermented and aged to and processed into a mildly sweet vinegar.

**Calcium Phosphate (CALPHOS) Animal bones (ruminants)** - the remain parts of the animals specifically the bone.

**Egg Shell** - the thin, hard outer layer of an egg, especially a hen's egg.

**Kuhol Shell** - serves for muscle attachment and calcium storage.
**Sea Shell** - the outer covering of a marine mollusk.

**Beneficial Microorganism (BMO)/ Indigenous Microorganism (IMO) Cooked Rice** - grains from palay that heated and serve as food.

**Lactic Acid Bacteria Serum (LABS) Fresh Milk** - an extract from ruminants.

**Powdered Rice** - a pulverized material out of rice.

## COMMON TOOLS & EQUIPMENT

**Chopping Board** - made up of plastic or wood used for slicing or cutting of the raw materials in preparing fermented concoction.

**Knife** - an aluminum metal used to cut or slice the raw materials in preparing fermented concoction.

**Manila paper** - a yellowish paper used to cover the pail with prepared fermented concoction.

**Marker Pen** - used in marking or labeling the output.

**Masking Tape** - used in fastening the Manila paper into the pail with prepared fermented concoction.

**Plastic Pail -** used as container of the prepared fermented concoction.

**Scissors/ cutter** - used in cutting the Manila paper and masking tape to make a presentable out-put.

**Stone** - a solid materials weighing at least 0.5 kg serve as stopper of the screen inside the pail

**Strainer / Screen** - made up of nylon or plastic that serve as filter inside the pail with prepared concoction.

**Weighing Scale** - a digital or manual tools used to measure mass of the raw materials in preparing fermented concoction.

# SELF CHECK 4.1-3
# Tools, Materials and Equipment

**Identification:**

1. Made up of plastic or wood used for slicing or cutting of the raw materials in preparing fermented concoction.

2. It is made up of rubber or plastic used to cover the hand to minimize the contamination of the fermented concoction.

3. Yellowish paper used to cover the pail with prepared fermented concoction.

4. A digital or manual tools used to measure mass of the raw materials in preparing fermented concoction.

4. made up of cloth or thin plastic used to cover the mouth eliminate contamination of the fermented concoction.

# ANSWER KEY 4.1-3

## Tools, Materials and Equipment

1. Chopping Board
2. Gloves
3. Manila paper
4. Weighing Scale
5. Face Mask

## JOB SHEET 4.1-3

## Tools, Materials and Equipment

| |
|---|
| **Title: Using tools, material and equipment in concoction process** |
| **Performance Objective:** At the end of this module the student should be able to use clean tools, materials and equipment in concoction process. |
| **Supplies/Materials:** Alugbati - Bamboo Shoots ,Banana Trunks ,Kamote Tops Kangkong.,Molasses, Mascuvado ,Brown sugar ,Banana, Squash, Papaya, Ampalaya Tomato, Fish, Golden Kuhol Meat ,Garlic Ginger ,Makabuhay ,Animal bones,Egg Shell ,Kuhol Shell , Sea Shell , Cooked Rice ,Fresh Milk - , Powdered Rice, Chopping Board , Knife, Manila paper, Marker Pen, Masking Tape. Plastic Pail , Scissors/ cutter , Stone,Strainer / Screen, Weighing Scale |
| **Steps/Procedures:**<br><br>Wear the PPE<br>Identify the tools, materials and equipment<br>Check if they have damage<br>Clean all the materials, tools and equipment<br>kept into the proper storage area<br>Put sign for identification<br>Do housekeeping |
| **Assessment Method:**<br><br>Demonstration |

## PERFORMANCE CHECKLIST 4.1-3
## Tools, Materials and Equipment

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee… | | |
| 1. Wear the PPE? | | |
| 2. Identify the tools, materials and equipment? | | |
| 3. Check if they have damage? | | |
| 4. Clean all the materials, tools and equipment? | | |
| 5. Kept into the proper storage area? | | |
| 6. Put sign for identification? | | |
| 7. Do housekeeping? | | |

**INFORMATION SHEET 4.1-4**
**Personal Hygiene**

# Introduction

Organic Agriculture is an ecological production management system that promotes and enhances biodiversity, biological cycles, and soil biological activity. It is based on minimal use of off-farm inputs and on management practices that restore, maintain, or enhance ecological harmony. Preparing tools and equipment must be in accordance with enterprise procedure and/ or aligned in the Philippine National Standard. The following must be observed:

1. Use of PPE or Personal Protective Equipment;
2. Use and proper handling of the appropriate tools, equipment and raw materials; and
3. Application of the 5 S of housekeeping. Unnecessary accident cause by hazardous activity in relation to preparation of organic concoction may lessen if not totally eliminated through the proper use of PPE. Appropriate handling of raw materials like; cleaning, sanitizing and securing work and storage areas needs the application of the 5 S of housekeeping.

**1. SEIRI** – means SORT by removing all unnecessary or not needed in the workplace to avoid accidents.

**2. SEITON** – means SET IN ORDER or SYSTEMATIZED by arranging all needed tools and equipment and raw materials to be used and properly labeled for easily identified.

**3. SEISO** – means SHINE or SWEEP by keeping or maintain our work area, tools, equipment and raw materials clean to avoid contaminated results.

**4. SEIKETSU** – means STANDARDIZE or SANITIZE by creating consistent way that task and procedure are done. Organic Agriculture Production Producing Organic Concoction and Exacts

**5. SHITSUKE** – means SELF-DISCIPLINE or SUSTAIN by making a habit of properly maintaining the correct procedure for this is the pillar of the first 4 S.

# SELF CHECK 4.1-4

## Personal Hygiene

**Identification**

1. Means SORT by removing all unnecessary or not needed in the workplace to avoid accidents.

2. means SET IN ORDER or SYSTEMATIZED by arranging all needed tools and equipment and raw materials to be used and properly labeled for easily identified.

3. Means SHINE or SWEEP by keeping or maintain our work area, tools, equipment and raw materials clean to avoid contaminated results.

4. Means STANDARDIZE or SANITIZE by creating consistent way that task and procedure are done. Organic Agriculture Production Producing Organic Concoction and Exacts

5. Means SELF-DISCIPLINE or SUSTAIN by making a habit of properly maintaining the correct procedure for this is the pillar of the first 4 S.

# ANSWER KEY 4.1-4

## Personal Hygiene

1. SEIRI
2. SEITON
3. SEISO
4. SEIKETSU
5. SHITSUKE

# JOB SHEET 4.1-4

## Personal Hygiene

| |
|---|
| **Title: Observing Personal Hygiene** |
| **Performance Objective:** At the end of this module you should be able to observe personal hygiene**.** |
| **Supplies/Materials:** PPE, materials. Tools, sanitize container and bottles Concoction area. |
| **Steps/Procedures:**<br><br>Wear the PPE.<br><br>Check the tools, materials in proper area<br><br>Arrange all needed tools and equipment and raw materials to be used and properly label for easily identify<br><br>Keep or maintain the work area, tools, equipment and raw materials clean to avoid contamination results<br><br>Create consistent way that task and procedure are done.<br><br>Make a habit of properly maintaining the correct procedure<br><br>Do housekeeping. |
| **Assessment method:**<br><br>**Demonstration:** |

# PERFORMANCE CHECKLIST 4.1-4

## Proper Personal Hygiene

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee.. | | |
| Wear the PPE? | | |
| Check the tools, materials in proper area? | | |
| Arrange all needed tools and equipment and raw materials to be used and properly label for easily identify? | | |
| Keep or maintain the work area, tools, equipment and raw materials clean to avoid contamination results? | | |
| Create consistent way that task and procedure are done? | | |
| Make a habit of properly maintaining the correct procedure | | |
| Do housekeeping? | | |

## Learning Outcome #2 Process of Concoction

**Assessment Criteria:**

Raw materials are prepared in accordance with enterprise practice.
Fermentation period is set based on enterprise practice.
Various concoctions are fermented following to organic practices.
Concoctions are harvested based on fermentation period of the concoction.

**Contents:**

Prepare raw materials

Fermentation period

Various concoctions

Period of harvest

**Tools Materials and Equipment**

Storage area, kangkong, camote tops, alugbati, malunggay, banana trunks, bamboo shoots and other fast growing green plants, Molasse/mascuvado/ brown sugar Ripe and sweet fruits but not limited to banana, papaya, watermelon, ampalaya , tomato Trash Fish and gills, scales, offal of big fishes, golden kuhol, meat, Garlic, ginger, Pure coconut vinegar animal bones, egg shell, sea shell, kuhol shell 1 kl. Cooked, cool rice900 ml. fresh milk, 100 ml clear liquid from fermented rice, Plastic pail, Wooden ladle, Manila paper or cheese cloth, String or rubber bands, Weighing scale, chopping board, Knife Marker, Strainer or nylon screen,

**References:**

https://www.ctahr.hawaii.edu/oc/freepubs/pdf/SA-8.pdf / (2) Making Culturing Lactic Acid Bacteria (LAB) http://www.cgnfindia.com/lab.html / (3) Lactobacillus Serum http://theunconventionalfarmer.com/recipes/lactobacillus-serum/

https://businessdiary.com.ph/3470/how-to-make-fermented-fruit-juice-or-ff

## LEARNING EXPERIENCES/ACTIVITIES

## LO # 2 Process of Concoction

| Learning Activities | Special Instructions |
|---|---|
| Read the attached Information Sheet 4.2-1 Prepare raw materials | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 4.2-1 on Prepare raw materials | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 4.2-1 on Prepare raw materials | |
| Perform Job Sheet 4.1-1 on Prepare raw materials | Observe the demonstration of your trainer and check the procedures while performing the operation. |
| Evaluate performance using performance checklist on 4.2-1 on Prepare raw materials | |
| Read the attached Information Sheet 4.2-2 on Fermentation period | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 4.2-2 on Fermentation period | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 4.2-2 on Fermentation period | |
| Perform Job Sheet 4.2-2 on Fermentation period | Observe the demonstration of your trainer and check the procedures while performing the operation. |
| Evaluate performance using performance checklist on 4.2-2 Fermentation period | |

| | |
|---|---|
| Read the attached Information Sheet 4.2-3 on Various concoctions | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 4.2-3 on Various concoctions | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 4.2-3 on Various concoctions | |
| Perform Job Sheet 4.2-3 on Various concoctions | Observe the demonstration of your trainer and check the procedures while performing the operation. |
| Evaluate performance using performance checklist on 4.2-3 on Various concoctions | |
| Read the attached Information Sheet 4.2-4 on Period of harvest | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 4.2-4 on Period of harvest. | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 4.2-4 on Period of harvest. | |
| Perform Job Sheet 4.2-4 on Period of harvest. | Observe the demonstration of your trainer and check the procedures while performing the operation. |
| Evaluate performance using performance checklist on 4.2-4 on Period of harvest. | |

***After doing this activity, you are now ready to proceed to the next module on*** *Package concoctions*

# INFORMATION SHEET 4.2-1
# Prepare raw materials

**Learning Objective:** After reading this information sheet, you should be able to prepare raw materials when producing organic concoction and extract.

**Introduction**

Preparation of raw materials is the process of preparing raw materials and chemicals. This is an important part because such preparation determines the basic constituents of paper. We manufacture products with various functions using our raw material preparation technologies, such as combination of different types of raw materials and blending of composite materials.

## Preparing raw materials

**FERMENTED PLANT JUICE** (FPJ) Natural Growth Enhancer, is **an artificial honey**. It is a nutritional activation enzyme and is very effective in natural farming.

**Materials:** kankong, Camote tops, banana trunk and molasses

**Choosing the raw materials for Fermented Plant Juice**
You choose materials that are:
Young and fresh
Free from insect pests and diseases
Abundant in the production area
Free from chemical containments

**Materials needed in making Fermented Plant Juice**
* Local plants that are fast growing like kangkong, legumes and grasses. You can also use bamboo shoots, asparagus shoots, actively growing plant parts and young fruits of cucumber, squash, melon, watermelon, ampalaya and other cucurbits.
* Weed species that are found growing in the production area, young leaves of trees, banana trunks, young leaves and fruits of stress tolerant crops are also good materials for Fermented Plant Juice.

* You can use either crude sugar or molasses or whichever is available and can be bought at a lower price.
* You will need basin, ceramic pot or plastic pail, net bag or cloth bag, paper or cloth for cover, string, stone as weight, bolo, chopping board, marking pen and glass jars.

**Steps in Making Fermented Plant Juice**

1. Collect the plant materials early in the morning while they are fresh and the microorganisms are still present. Do not wash the plant materials.

2. cut the plant materials into small pieces so that the juice can be easily extracted.

3. Put 3 kg chopped plant materials in a basin, add 1 kg crude sugar or molasses, then mix thoroughly with your hands. Make sure that all plant materials are mixed with sugar so that the juice can be extracted easily.

4. Put the mixture in a net bag or cloth bag. This is done so that the extracted juice will ooze from all sides of the bag.

5. Put the bagged mixture in a ceramic pot or plastic pail, and put weight to compress the mixture. Stone is a good material used to weigh down the mixture.

6. Cover the pot or pail with paper or cloth, and secure with a string or rubber band. Paper or cloth is used as cover to allow some air to get inside the pot and for the gas that is being produced during the fermentation process to escape. On the cover, write the date of processing and the expected date of harvest

7. Store the container with the bagged mixture in a cool dry shady place. Make sure that the storage area is not infested with cockroaches or mice, because they might feed on the mixture and contaminate the extract. In 7 days, plant juice is extracted and fermented. The plant extract will change its color from green to yellow, then to brown and will smell sweet and alcoholic.

8. After 7 days, lift the bagged mixture and squeeze hard to get the remaining extracts

9. Collect the fermented extracts and preserve in dark colored glass jar. To cover the jar, use paper or cloth to allow the gas to escape during further fermentation, then, store in a cool, shady place. You may add the plant residue to the compost

pile to hasten decomposition or you can apply it to the garden plots as source of organic matter. Use your Fermented Plant Juice more effectively if it is stored for another one week after completion.

**2. Fermented Fruit Juice (FFJ)** Natural Taste Enhancer **-** is made from sweet ripe fruits, fruit vegetables and root crops**.** Thoroughly blended with crude sugar or molasses and stored for a short period of time, the fermented extract is applied to the plants to promote flowering and fruit setting.

**Materials:** banana papaya kalabasa and Molasses

**Choosing the materials for Fermented Fruit Juice**
You must choose materials that are:
locally produced
free from insect pests and diseases
not fit for human consumption

**Materials needed in making Fermented Fruit Juice**

* Locally produced sweet ripe fruits like mango, banana, papaya, strawberry and chico; ripe squash fruit and matured carrot; and root crops particularly camote, cassava and gabi. Citrus fruits are not recommended. You can make Fermented Fruit Juice from single material or a combination of materials. The extract from the combination of banana, papaya, and squash have been proven to be effective in flower induction and fruit setting by many organic farmers.
* You can use either crude sugar or molasses or whichever is available or can be purchased at lower cost.
* You will also need ceramic pots or plastic pail, basin, net bag or cloth bag, paper or cloth for cover, string, stone as weight, bolo, chopping board, marking pen, and glass jars for storage.

**Procedure**

1. Collect ripe fruits or vegetables that are already available or in season, for example, if squash is available, then make fermented squash juice. There are plenty of materials to be used so you can make different kinds of Fermented Fruit Juice. Use any materials that are free from insect pests and diseases.

2. Chop the materials into small pieces so that the juice can be easily extracted.

3. Put 1 kg chopped materials in a basin, add 1 kg crude sugar or molasses, and then mix thoroughly with your bare hands. You must make sure that all chopped

materials are coated with sugar or molasses so that the juice can be extracted easily.

4. Put the mixture in a net bag or cloth bag. This is done so that the extracted juice will ooze from all sides of the bag. Put the bagged mixture in a ceramic pot or plastic pail, and put weight to compress the mixture. Stone is a good material used to weigh down the mixture.

5. Cover the pot or pail with paper or cloth and secure with a string or rubber band. Paper or cloth is used as cover to allow some air to get inside the pot or pail and for the gas that is being produced during the fermentation process to escape. On the cover, write the date of processing and the expected date of harvest.

6. Store the container with the bagged mixture for 7 days in a cool dry shady place. Make sure that the storage area is not infested with cockroaches or mice, because they might feed on the mixture and contaminate the extract. In 7 days, plant juice is extracted and fermented. The fruit extract will change its color from yellow orange to brown, and will smell sweet and alcoholic. After 7 days, lift the bagged mixture and squeeze hard to get the remaining extracts.

7. Collect the fermented extracts and preserve in dark colored glass jar. To cover the jar, use paper or cloth to allow the gas to escape during further fermentation, then, store in a cool, shady place. You may add the fruit residue to compost pile to hasten decomposition or you can apply it to the garden plots as source of organic matter. You can use your Fermented Fruit Juice more effectively if it is stored for another one week after completion.

**Oriental Herbal Nutrients** (OHN) Natural Pesticides- OHN is a mixture of edible, aromatic herbs extracted with alcohol and fermented with brown sugar. It is used to discourage the growth of anaerobic, potentially pathogenic microbes and encourage beneficial aerobic microbes in the soil and on plants. Herbs long recognized by many ancient cultures as having such prebiotic properties include fresh ginger root (Zingiber officinale),

**Materials:** 1 garlic 1 kg. Ginger 200 grms. Mascuvado sugar 2.2 liters pure coconut vinegar

**A. Preparation of Fresh Herb Extracts** (when using fresh ginger or turmeric root and garlic cloves)

1. Slice or crush fresh ginger OR turmeric root, weigh, and place in a clean glass jar to fill 2/3 full. Slice or crush garlic cloves

2. Add an equal amount of brown sugar by weight to each jar. Cover the jars with muslin or a paper towel and secure with a rubber band or threaded ring portion of a Mason jar and let sit for 5 to 7 days at room temperature out of direct sunlight.

3. Fill each jar with vodka (or other liquor that is 40% proof).

4. Replace the jar's cover. Let sit at room temperature, stirring clockwise with a wooden spoon every morning for 14 days.

5. Strain 1 /3 of the liquid from each jar into separate, labeled glass jars ("Ginger OR Turmeric Extract" and "Garlic Extract," respectively).

6. Repeat Steps 3 through 5, adding to the respective extract jars.

7. This extraction process (Steps 3 through 6) can be repeated up to 5 times before discarding the herb, brown sugar, and liquor mixtures (which can be composted or made into tea).

**Indigenous Microorganisms** (IMO) Beneficial Microorganisms Indigenous Microorganism (IMO) In natural farming, Indigenous Microorganism (IMO) is becoming popular among farmers. This Indigenous microorganism (IMO) has been successfully tried by government agriculturists, academic researchers, non-profit organizations and farmers alike. They have found that IMO is useful in removing bad odors from animal wastes, hastening composting, and contributing to crops' general health.

**A. Materials**

1. Clay pot/Bamboo trough

2. Manila paper (unprinted)

3. Basin

4. Cooked rice

5. Muscovado sugar (generic or crude sugar)

6. Clean water (no chlorine or other chemicals)

**How to Make Your Own Indigenous Microorganism (IMO):**

1. Cook a kilo of rice, preferably organic. After cooling, put the cooked rice in a wooden, earthen or ceramic container. Avoid plastic or aluminum.
2. Cover the mouth of the container completely with cloth or paper, fixed in place with a rubber band, to prevent water or small insects from getting in.

3. Put the covered container, protected from possible rain, under the trees, in a bamboo grove, a forest floor, or wherever a thick mat of leaves has formed. Leave it there for three days.

4. After whitish moldy filaments have formed, transfer the entire contents of the container to a larger glass or earthen jar and add one kilo of brown sugar or molasses, preferably organic.

5. Cover the jar with clean cloth or paper, fixed with a rubber band. Keep the jar in a dark, cool place. Let it ferment for seven days, until it appears muddy. This is your IMO concoction.

**Procedure**

**1. Collecting IMO**

a. Place cooked rice into pot or bamboo trough. Let it cool first before placing into the trough.

b. Cover container with fine wire screen to avoid rat disturbances and tie up using any tying material.

c. Place container face down or slant position in an area where decomposed crops such as corn, rice straw, etc. or in banana/bamboo plantation areas. Cover container with any material to protect from rain.

d. Collect container after 5-7 days when presence of molds can be seen.

**2. Culture and production**

a. Transfer the molded rice with collected microbes into a basin. For every kilo of cooked rice add 1 kilo of muscovado sugar and 1 liter clean water (no chlorine). Mix well.

b. Transfer the mixture into an old pail or clay jar. Cover with unprinted Manila paper and tie up using any tying materials. Fill the pail up to 75% only, leaving 25% air space.

c. Place the container in a cool place away from the heat of the sun. d. Leave pail or jar for 7 days then collect by straining the liquid extract, leaving the substrate to the compost area.

**Lactic Acid Bacteria Serum (LABS)** It can convert waste into organic matter and basic minerals, good digestion for animals Deodorizer. Lactic acid bacteria can be collected in the air.

**A. Materials**

1. Rice wash

2. Fresh milk (skimmed or powdered milk can be used)

3. Used or old pail or plastic container 4. Manila paper (unprinted)

5. Muscovado sugar (crude or generic sugar)

**B. Procedure**

1. Pour rice wash (solution generated when you wash the rice with water) into a container.

2. Allow 50-75% air space in the container.

3. Cover container loosely (not vacuum tight, allowing air to move into the container). Put container in a cool area with no direct sunlight.

4. Allow rice wash to ferment for 5-7 days at a temperature of 20-25 degrees centigrade.

5. The rice bran will be separated and float like a thin film on the liquid smelling sour.

6. Strain the liquid with a cheese cloth or wheat flour bag cloth. Put liquid in a bigger container.

7. Pour ten parts milk (the original liquid has already been infected with different types of microorganisms including Lactobacilli. Saturation of milk will eliminate the other microorganisms and pure Lactobacilli will remain.)

8. Ferment in 5-7 days. Carbohydrates, protein and fat will

**Calphos Natural calphos** micronutrients flower inducer, prevent overgrowth

**Materials**

1. Any of the following: pork, fish and beef bones, eggshells and kuhol and/or any shells

2. Clay pot or cross-cut bamboo trough

3. Manila paper (unprinted)

4. Plastic straw (for tying)

5. Coconut vinegar

6. Griller

**B. Procedure**

1. Broil bones. Roast eggshells until they turned into ashes

2. Pulverize bones. Transfer in a container pulverized bones or shells and add equal volume of vinegar.

3. Transfer the mixture into a bamboo trough or clay jar, cover with Manila paper and tie up with plastic straw.

4. for bones: Allow to sit for one month or until bones soften or dissolve completely. For eggshells: Allow to sit for 2 weeks (14 days) or until dissolved completely.

5. Harvest, strain the preparation and bottle after a month or until when bones are completely dissolved.

**Fish Amino Acid (FAA)** is a liquid made from fish scrap. FAA is of great value to both plants and microorganisms in their growth, because it contains and abundant amount of nutrients and various types of amino acids that will constitute a source of Nitrogen for plants.

**Materials**

1. Chopped fish or fish trash such as gills, entrails, golden snail (shell removed) or meat scrap and rejects.

2. Old or used pail

3. Manila paper (unprinted)

4. Muscovado sugar

5. Plastic straw

6. Clean water (no chlorine or other chemical compound)

**B. Procedure**

1. Mix properly the following ingredients at a ratio of 1:1:1 3 kilos chopped fish, snail or meat scraps and rejects 3 kilos muscovado sugar

2. Plants that are fast growing like kangkong, legumes and grasses. You can also use bamboo shoots, asparagus shoots, actively growing plant parts

## SELF CHECK 4.2-1

## Prepare raw materials

**Identification**

1. A liquid made from fish scrap. Abundant amount of nutrients and various types of amino acids that will constitute a source of Nitrogen for plants.

2. Plants that are fast growing like kangkong, legumes and grasses. You can also use bamboo shoots, asparagus shoots, actively growing plant parts

3. Made from sweet ripe fruits, fruit vegetables and root crops.

4. useful in removing bad odors from animal wastes, hastening composting, and contributing to crops' general health.

5. Is a mixture of edible, aromatic herbs extracted with alcohol and fermented with brown sugar. It is used to discourage the growth of anaerobic, potentially pathogenic microbes and encourage beneficial aerobic microbes in the soil and on plants

## ANSWER KEY 4.2-1

## Prepare raw materials

### Identification

1. Fish Amino Acid (FAA)
2. Fermented plant juice
3. Fermented Fruit Juice
4. IMO
5. OHN

# JOB SHEET 4.2-1

## Prepare raw materials

| |
|---|
| **Title: Preparing raw materials for producing organic concoction and extract.** |
| **Performance Objective:** At the end of this module the student should be able to prepare raw materials when producing organic concoction and extract. |
| **Supplies/Materials:** kankong, Camote tops, banana trunk and molasses, banana papaya kalabasa, garlic, ginger, Mascuvado, sugar pure ,coconut vinegar, Clay pot/Bamboo trough, Manila paper, Basin, Cooked rice, Rice wash, Fresh milk (skimmed or powdered milk can be used), pork, fish and beef bones, eggshells and kuhol and/or any shells. |
| **Steps/Procedures:**<br><br>Wear PPE<br>Collect all the raw materials<br>Eliminate the unnecessary and unwanted one<br>Make sure that all raw materials are clean<br>Check the raw materials are suited to the types of concoctions<br>Do housekeeping |
| **Assessment Method:**<br><br>Demonstration |

# PERFORMANCE CHECKLIST 4.2-1

## Prepare raw materials

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee… | | |
| 1. Wear PPE? | | |
| 2. Collect all the raw materials? | | |
| 3. Eliminate the unnecessary and unwanted one? | | |
| 4. Make sure that all raw materials are clean? | | |
| 5. Check the raw materials are suited to the types of concoctions? | | |
| 6 Do housekeeping? | | |

## INFORMATION SHEET 4.2-2
## Fermentation period

**Learning Objective:** After reading this information sheet, you should be able to determine the period of fermentation process.

**Introduction:**

Fermentation refers to the metabolic process by which organic molecules (normally glucose) are converted into acids, gases, or alcohol in the absence of oxygen or any electron transport chain. Fermentation pathways regenerate the coenzyme nicotinamide adenine dinucleotide (NAD+), which is used in glycolysis to release energy in the form of adenosine triphosphate (ATP). Fermentation only yields a net of 2 ATP per glucose molecule (through glycolysis), while aerobic respiration yields as many as 32 molecules of ATP per glucose molecule with the aid of the electron transport chain.

The study of fermentation and its practical uses is named zymology and originated in 1856 when French chemist Louis Pasteur demonstrated that fermentation was caused by yeast. Fermentation occurs in certain types of bacteria and fungi that require an oxygen-free environment to live (known as obligate anaerobes), in facultative anaerobes such as yeast, and also in muscle cells when oxygen is in short supply (as in strenuous exercise). The processes of fermentation are valuable to the food and beverage industries, with the conversion of sugars into ethanol used to produce alcoholic beverages, the release of $CO_2$ by yeast used in the leavening of bread, and with the production of organic acids to preserve and flavor vegetables and dairy products.

## Function of Fermentation

The main function of fermentation is to convert NADH back into the coenzyme NAD+ so that it can be used again for glycolysis. During fermentation, an organic electron acceptor (such as pyruvate or acetaldehyde) reacts with NADH to form NAD+, generating products such as carbon dioxide and ethanol (ethanol fermentation) or lactate (lactic acid fermentation) in the process.

## Types of Fermentation

There are many types of fermentation that are distinguished by the end products formed from pyruvate or its derivatives. The two fermentations most commonly used by humans to produce commercial foods are ethanol fermentation (used in beer and bread) and lactic acid fermentation (used to flavor and preserve dairy and vegetables).

## Ethanol Fermentation

This figure depicts the processes of glycolysis and ethanol fermentation. In ethanol fermentation, the pyruvate produced through glycolysis is converted to ethanol and carbon dioxide in two steps. First, the pyruvate releases carbon dioxide to form a two-carbon compound called acetaldehyde. Next, acetaldehyde is reduced by NADH to ethanol, hereby regenerating the NAD+ for use in glycolysis. Overall, one molecule of glucose is converted into two molecules of carbon dioxide and two molecules of ethanol. Ethanol fermentation is typically performed by yeast, which is a unicellular fungus.

## Lactic Acid Fermentation

This figure depicts the processes of glycolysis and homolactic fermentation. There are two main types of lactic acid fermentation: homolactic and heterolactic. Inhomolactic acid fermentation, NADH reduces pyruvate directly to form lactate. This process does not release gas. Overall, one molecule of glucose is converted into two molecules of lactate. In heterolactic fermentation, some lactate is further metabolized, resulting in ethanol and carbon dioxide via the phosphoketolase pathway. Lactic acid fermentation is primarily performed by certain types of bacteria and fungi. However, this type of fermentation also occurs in muscle cells to produce ATP when the oxygen supply has been depleted during strenuous exercise and aerobic respiration is not possible.

**1. Fermented Plant Juice**
**Materials**: 1 kg kankong;1 kg Camote tops; 1 kg banana trunk; and 1.5 kgs molasses

**Steps on to ferment:**
1. Clean and wash vegetable materials
2. Drain for 5 minutes
3. Slice to an inc size
4. Mix all vegetable thoroughly in a plastic pail(20 li. Capacity)
5. Mix with 1.5 kgs of molasses thoroughly
6. Put nylon screen on top of the mixture
7. Put 5-8 pieces 25-30 grams stone on top of the nylon screen
8. 8. Wipe the mouth of the plastic pail
9. Cover with two layered manila paper
10. Tie with rubber band
11. Put marking on the masking tape bearing the name and date of fermentation and paste it on top of the manila paper
12. Keep in dark cool room for 7 days
13. Open the mixture and extract the liquid
14. Filter the liquid and keep it in a plastic container (do not close the cap tightly, loosen the cap of approximately 1 complete twist)
15. Completely close cap after a week or when there are no bubbles going up
16. The concoction is ready to use after extraction

**2. Fermented Fruit Juice (FFJ) Extracts**
**Materials:** 1 kg banana fruit; 1kg papaya; 1kg watermelon; and 3 kgs molasses
Steps on how to ferment:

1. Clean and wash fruits
2. Drain for 5 minutes
3. Slice to an inch size
4. Mix all fruits thoroughly in a plastic pail ( 20 liters capacity)
5. Mix with 3 kgs. Of molasses thoroughly
6. Put nylon screen on top of the mixture
7. Put 5-8 pieces 25-30 grams stone on top of the nylon screen
8. Wipe the mouth of the plastic pail
9. Cover with two layered manila paper
10. Tie with rubber band
11. Put marking on the masking tape bearing the name and date of fermentation

and paste it on top of the manila paper
12. Keep in dark cool room for 7 days
13. Open the mixture and extract the liquid
14. Filter the liquid and keep it in a plastic container ( do not close the cap tightly, loosen the cap of approximately 1 complete twist)
15. Completely close cap after a week or when there are no bubbles going up
16. The concoction is ready to use after extraction

**4. Fish Amino Acid ( FAA)**
**Materials :** 1 kg Trash fish and gills, scales, offals, of big fish; and 1 kg molasses

**Steps on how to ferment:**
1. Clean and wash vegetable materials
2. Drain for 5 minutes
3. Slice to an inch size
4. Mix all parts thoroughly in a plastic pail ( 20 liters capacity)
5. Mix with 1 kgs of molasses thoroughly
6. Put nylon screen on top of the mixture
7. Put 5-8 pieces 25-30 grams stone on top of the nylon screen
8. Wipe the mouth of the plastic pail
9. Cover with two layered manila paper
10. Tie with rubber band
11. Put marking on the masking tape bearing the name and date of fermentation and paste it on top of the manila paper
12. Keep in dark cool room for 15 days
13. Open the mixture and extract the liquid
14. Filter the liquid and keep it in a plastic container (do not close the cap tightly, loosen the cap of approximately 1 complete twist)
15. Completely close cap after a week or when there are no bubbles going up
16. The concoction is ready to use after extraction

**5. Oriental Herbs Nutrients (OHN)**
Materials: 1 kg garlic; 1kg. Ginger, 200 grams muscuvado, and 2.2 liters pure coconut
vinegar

**Steps on how to ferment:**
1. Skinned the garlic and ginger
2. Cut garlic in halves and slice ginger into quarter of an inch
3. Mix garlic and ginger with 200 grams muscuvado sugar in a plastic pail
4. Wipe the mouth of the plastic pail

5. Close cover tightly and seal it with masking tape at the side
6. Mark the name and date of fermentation (First stage fermentation)
7. Open the cover and add 2.2 liters of pure coconut vinegar (1:1 solution) 3 days after
8. Wipe the mouth of the plastic pail
9. Close cover tightly and seal it again with masking tape
10. Open the cover tightly and decant the liquid to another container 10 days after (second stage fermentation)
11. Close tightly the cover and do the markings ( first extraction)
12. The concoction is now ready to use for animals
13. Add the same amount of liquid extracted ( 2.2 liters of pure coco vinegar) then add 200 grams of hot pepper and 100 grams of panyawan, soak for 10 days ( secondextraction)
14. Repeat number 13 procedure for the third extraction

6. Indigenous Micro-organism (IMO)
**Materials:** 1 kg commercial rice; and 2 kgs molasses

**Steps on how to ferment:**
1. Wash the rice properly. (Keep the first wash liquid for the LABS)
2. Cook it normally (not too wet or too dry)
3. Cool the cooked rice naturally
4. Transfer the cooked rice to a tray
5. Use wooden ladle to transfer rice
6. Put some cooked rice inside the bamboo pole (1/4 full of rice)
7. Cover it with a two layered manila paper then tie with rubber bands
8. Wrap the bamboo pole with a clean cellophane then tie with rubber bands
9. Write markings on the masking tape bearing the name and date of fermentation and
paste it on top of the cellophane then tie with rubber bands
10. Keep it under the bamboo forest for 3 to 5 days
11. Open the bamboo pole and inspect the growing molds, black colored molds discard, white colored molds collect
12. Weigh the recovered rice and molds, and add molasses in equal weight
13. Put the mixture in a plastic container, wipe the mouth, cover with a double layered manila paper and put the proper markings
14. Drain the Liquid from the mixture, filter and place it in another container (do not close the cap tightly; loosen the cap of approximately 1 complete twist)
15. Completely close cap after a week or when there are no bubbles going up
16. The concoction is ready to use after extraction.

**7. Lactic Acid Bacteria Serum (LABS)**

**Materials:** 900 ml. Cow's milk; 100 ml. Clear liquid from fermented rice; and 1 liter molasses

**Steps on how to ferment:**
1. Use the first wash liquid from the cooked rice
2. Put liquid inside the plastic container (3/4 full) and wipe excess water
3. Cover the container with a double layered manila paper
4. Mark the name and date of fermentation
5. Ferment it for 7 days (first stage fermentation)
6. Use 1 liter cow's milk pack and remove 100 ml (10%)
7. Extract 100ml from the fermented first wash liquid of the cooked rice
8. Take the liquid between the bottom and top layers of the fermentation and add to the 1 liter milk pack
9. Return the cover of the pack and seal it with a masking tape
10. Mark it with the name and date of fermentation
11. Keep it for 5 days in a dark cool room, do nt disturb
12. Drain the liquid (whey) and filter, separate the sludge from the liquid
13. Measure the liquid and add the same amount of molasses
14. Keep it in a plastic container (do not close the cap0 tightly, loosen the cap of approximately 1 complete twist)
15. Completely close cap after a week or when there are no bubbles going up.
16. The concoction is ready to use after extraction.

**8. Calphos Micro Nutrients (CALPHOS)**
**Materials:** 3 kgs. Cow bones/eggshell and 27 liters of pure coco vinegar

**Steps on how to ferment:**
1. Clean and wash cow bones properly
2. Cook bones with some flesh and spices
3. Remove all meat and fats thoroughly (eat the meat and fats)
4. Wash and clean the bones
5. Put the bones above fired charcoal
6. Wait until the remaining fats are drained
7. Remove the bones when it became brownish in color (do not make it black, it's over cooked)
8. Cool it off for 10 to 20 minutes
9. Wash again to remove excess oil, if necessary
10. Drain excess water
11. Put the bones inside the plastic pail
12. Add 27 liters of pure coconut vinegar without coloring
13. Wipe the mouth of the plastic pail, cover with two layered manila paper and

write the markings (name and date of fermentation)
14. Open the container after 30 days of soaking, filter the liquid and keep it in another plastic container (do not close the cap tightly, loosen the cap of approximately 1
complete twist)
15. Completely close cap after a week or when there are no bubbles going up. Write the proper markings. It is ready to use.

**8. Natural Human Health Enhancer ( 3Cs)**
**Materials:** 5 kgs. Celery; 5 kgs. Carrots; 5 kgs. Cucumber; and 5 Kgs. Muscuvado

**Steps on How to ferment:**
1. Clean and wash vegetable materials thoroughly
2. Drain for 5 minutes
3. Slice to ¼ of an inch size
4. Mix all vegetable thoroughly in a plastic container (80 Liters capacity)
5. Mix with 5 kgs of muscuvado thoroughly
6. Put nylon screen on top of the mixture
7. Put 6-8 pieces of 250-500 ml mineral water container on top of the nylon screen as weight
8. Wipe the mouth of the plastic pail
9. Cover with two layered Manila paper
10. Tie with rubber band
11. Write markings on the masking tape bearing the name and date of fermentation and paste it on top of the Manila paper.
12. Keep in dark cool room for 20 days
13. Open the mixture and extract the liquid
14. Filter the liquid and keep it in a plastic container (do not close the cap tightly, loosen the cap of approximately 1 complete twist)
15. Completely close cap after a week or when there are no bubbles going up
16. The concoction is ready to use after extraction

## SELF CHECK 4.2-2
## Fermentation period

### True or False

1. Fermentation is the metabolic process by which organic molecules (normally glucose) are converted into acids, gases, or alcohol in the absence of oxygen or any electron transport chain.

2. French chemist Louis Pasteur demonstrated that fermentation was caused by yeast?

3. Filter the liquid and keep it in a plastic container (do not close the cap tightly, loosen the cap of approximately 1 complete twist)

4. Lactic acid fermentation is primarily performed by certain types of bacteria and fungi

5. The processes of fermentation are valuable to the food and beverage industries, with the conversion of sugars into ethanol used to produce alcoholic beverages, the release of $CO_2$ by yeast used in the leavening of bread, and with the production of organic acids to preserve and flavor vegetables and dairy products.

## ANSWER KEY 4.2-2

## Fermentation period

**True or False**

1. True
2. True
3. True
4. True
5. true

# JOB SHEET 4.2-2
# Fermentation period

| |
|---|
| **Title:** Determining period of fermentation process.. |
| **Performance Objective:** At the end of this module the student should be able to determine the period of fermentation process. |
| **Supplies/Materials:** PPE, Various concoctions (FPJ, FFJ, IMO, FAA, LABS, CALPHOS) |
| **Steps/Procedures:**<br><br>Wear the PPE<br><br>Put the label in the fermented concoctions<br><br>Keep the concoction in the shaded area until undergone fermentation process<br><br>Do housekeeping |
| **Assessment Method:**<br><br>Demonstration |

## PERFORMANCE CHECKLIST 4.2-2

## Fermentation period

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee… | | |
| 1. Wear the PPE? | | |
| 2.Put the label in the fermented concoctions | | |
| 3. Keep the concoction in the shaded are until undergone fermentation process | | |
| 4 Do housekeeping? | | |

# INFORMATION SHEET 4.2-3
# Various concoctions

**Learning Objective:** After reading this information sheet, you should be able to identify various type of concoctions.

## INTRODUCTION

The insurmountable rising cost of inorganic fertilizers inevitably uncontrollable in the coming production years. Looking into this perspective the farmers has to look for an alternative measure to sustain his farming business profitability. Here comes the discovery of using concoctions. Concoctions is a combination of various ingredients, usually herbs, spices, condiments, powdery substances or minerals, mixed up together, minced, dissolved or macerated into a liquid so as they can be ingested or drunk. The term "concoction" is sometimes loosely used metaphorically in order to describe a cocktail or a motley assemblage of things, persons or ideas. Various concoctions:

1. **Indigenous Microorganism (IMO)** Indigenous microorganisms are a group of innate microbial consortium that inhabits the soil and the surfaces of all living things inside and out which have the potentiality in biodegradation, nitrogen fixation, improving soil fertility, phosphate solubilizes and plant growth promoters.

**USES**

- Serve as Foliar Spray, applied to the soil-one 1day before direct seedling or transplanting

**PURPOSE**

- Hasten decomposition of compost and increase soil fertility
- Serve as foliar fertilizer and good soil conditioner
- Restore plant vitality
- Reduces growth of weeds and grasses seeds
- Speed up plant growth
- Revives nutrients in the soil

2. **Fermented Plant Juice (FPJ)** Fermented plant juice (FPJ) is a fermented extract of plants which helps crops to absorb nutrients directly for healthy growth and enabling their potential. It consist the young shoots of vigorously growing plants that are allowed to ferment for approximately 7 days with the aid of brown sugar. The brown sugar draws the juices out of the plant material via osmosis and also serves as a food source for the microbes carrying out the fermentation process. The weak alcohol produced during fermentation extracts chlorophyll (soluble in ethanol) and other plant components.


**USES**

- Growth hormone promotant
- Chlorophyll enhancer, microbe's fortifier. Spray as side dressing during " Vegetative Stage" (Pagpangipli)
- Apply on spray during " change Over Period" (pamusog-busog) and Reproductive Stage (pamuging)

**PURPOSE**

- Food for Microorganism
- Enhances plant growth
- Greening of leaves and produces chlorophyll
- Makes plant strong and builds resistance against pests and diseases

**3. Fermented Fruit Juice or FFJ-** is made from sweet ripe fruits, fruit vegetables and root crops. Thoroughly blended with crude sugar or molasses and stored for a short period of time, the fermented extract is applied to the plants to promote flowering and fruit setting.


**USES**

- Good source of potash (Potassium)

**PURPOSE**

- Sweetens the fruit and serves as source of potassium which can speed up plant absorption and result to sweeter tasting fruits

- Increases plant nutrition through leaves and roots with potassium factor
- Improves plant growth an soil health
- Acts as plant hormone and growth promotant
- Speeds up harvestings


**4. Fish Amino Acid (FAA**) the Fish Amino Acid (FAA) is a liquid made from fish. FAA is of great value to both plants and microorganisms in their growth, because it contains and abundant amount of nutrients and various types of amino acids (will constitute a source of nitrogen (N) for plants). ... Fish trash (head, bone, intestine, etc.)

**USES**

- Nitrogen Fixing (Urea Fertilizer)
- Nitrate from fish contains
- Abundant amount of nutrient and various type of amino acid

**PURPOSE**

- Serve as nitrogen fertilizer, good soil conditioner
- Rich in calcium and protein and rich supplement for plants
- Serve as “growth hormone” for plant growth and development

Food for microorganism


**5. Oriental Herbal Nutrients (**OHN) OHN is a mixture of edible, aromatic herbs extracted with alcohol and fermented with brown sugar. It is used to discourage the growth of anaerobic, potentially pathogenic microbes and encourage beneficial aerobic microbes in the soil and on plants.

**USES**

- For rice, corn vegetables and banana.
  Spray from planting up to bearing stage

**BENEFITS**

- Natural pest repellent. It is use throughout the early, vegetative, change over and fruiting stages
- It is very important in natural farming

**PURPOSE**

- Natural pesticide and fungicide
- Serve as pesticide and fungicide for plants and animals
- Repel insect
- Prevent plant pest and diseases
- Promote healthy immune system
- Treatment of fungal problem of plants, downy mildew, powdery mildew

6. **Calcium Phosphate (Cal Phos**) Calcium phosphate is a family of materials and minerals containing calcium ions ($Ca^{2+}$) together with inorganic phosphate anions. Some so-called calcium phosphates contain oxide and hydroxide as well. They are white solids of nutritious value. Calcium phosphates occur abundantly in nature in several forms and are the principal minerals for the production of phosphate fertilizers and for a range of phosphorus compounds. For example, the tribasic variety (precipitated calcium phosphate), $Ca_3(PO_4)_2$, is the principal inorganic constituent of bone ash.

**USES**

- Induces flowering and prevent overgrowth of plants
- Strengthen flowers

**PURPOSE**

- Makes fruit hard and sweet
- Prevent overgrowth and applied when nitrogen is big
- Increases calcium factor on roots and leaves
- Applied when the plants are about to flower
- Induces flowering among plants
- Induces longer shelf life in fruits

**7. Lactic Acid Bacteria Serum (LABS**) Lactic Acid Bacteria (LAB) are widespread microorganisms which can be found in any environment rich mainly in carbohydrates, LAB are anaerobic microorganism that decompose sugar in the absence of oxygen. Normally they are separated and cultured with rice washed with water and milk. This is how Lactic Acid Bacteria Serum (LABS) obtained. Lactic Acid Bacteria Serum is now used for its ability to convert waste into organic matter and basic materials. And they thrive and feed on the ammonia released in the decomposition normally associated with foul odor (removes foul odor). Lastly they serve as defense against pathogenic diseases such as harmful fungi and viruses.


**USES**

- It converts waste into organic matter and basic minerals

**PURPOSE**

- Improve growth rate on plants and improve digestive on animals
- Prevent foul odor

## SELF CHECK 4.2-3

## Various concoctions

**Multiple choice.** Write the letter of your choice to your answer sheet.

1. They are microorganisms which can be found in any environment rich mainly in carbohydrates.
A. Fungi B. Viruses C. Nematodes D. Lactic acid bacteria

2. What is the real word for OHN?
A. Orange Herbal Nutrient B. Overload Herbal Nutrient C. Oriental Herbal Nutrient D. Official Herbal Nutrient

3. It occurs abundantly in nature in several forms and are the principal minerals for the production of phosphate fertilizers and for a range of phosphorus compounds.
A. Calcium phosphate B. Nitrogen Phosphate C. Hydrogen Phosphate D. None of the above

4. It promotes the plant produce more flower and fruits.
A. OHN B. FPJ C. FFJ D. Cal Phos

5. It is a fermented extract of plants which helps crops to absorb nutrients directly for healthy growth and enabling their potential.
A. OHN B. FFJ C. FPJ D. IMO

## ANSWER KEY 4.2-3

## Various concoctions

### Multiple choice

1. D
2. C
3. A
4. C
5. C

## JOB SHEET 4.2-3
## Various concoctions

| |
|---|
| **Title: The various concoctions** |
| **Performance Objective:** At the end of this module the student should be able to identify the different types of concoctions. |
| **Supplies/Materials:** PPE, various concoction |
| **Steps/Procedures:**<br><br>Wear the PPE.<br>Identify the types of concoctions<br>Determine the benefits and uses<br>Do housekeeping |
| **Assessment Method:**<br><br>Demonstration |

## PERFORMANCE CHECKLIST 4.2-3
## Various concoctions

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee... | | |
| 1. Wear the PPE? Wear the PPE? | | |
| 2 Identify the types of concoctions? | | |
| 3. Determine the benefits and uses? | | |
| 5. Do housekeeping? | | |

# INFORMATION SHEET 4.2-4
# Period of harvest

**Learning Objective:** After reading this information sheet, you should be able to identify harvesting time of concoction.

**Introduction:**

The knowledge, skills and attitude required to produce organic fertilizers which include tasks such as preparing composting area and raw materials and carrying-out composting activities and finally, harvesting of fertilizer. Using Organic Concoction is connecting with the nature. Embracing the mystery of the environment that God created. Significance of Organic Concoctions 1. Replaces the chemical based fertilizers, pesticides, fungicides, repellants, growth enhances and other food ingredients for animals and plants; 2. Safe for carcinogenic substance; 3. An environment-friendly; 4. Low cost; and 5. Many business opportunities

## PROCEDURE IN HARVESTING ORGANIC CONCOCTIONS

**Fermented Plant Juice (FPJ)**
After Fermenting for 7 days. Extract the liquid using strainer or a piece of cloth. Keep it in a plastic container. FPJ is ready to use.

**Fermented Fruit Juice (FFJ)**
After Fermenting for 7 days. Extract the liquid using strainer or a piece of cloth. Keep it in a plastic container. FFJ is ready to use.

**Fish Amino Acid (FAA)**
After Fermenting for 7 days. Extract the liquid using strainer or a piece of cloth. Keep it in a plastic container. FAA is ready to use.

**Oriental Herbal Nutrient (OHN)**
1. After Fermenting for 10 days.
2. Extract the liquid using strainer or a piece of cloth.
3. Keep it in a plastic container. Organic Agriculture Production Producing Organic Concoction and Exacts
4. OHN is ready to use. Calcium Phosphate (CALPHOS)

After Fermenting for 30 days. Extract the liquid using strainer or a piece of cloth. Keep it in a plastic container. CALPHOS is ready to use.

**Beneficial Microorganism (BMO)/Indigenous Microorganism (IMO)**
After Fermenting for 1 week. Extract the liquid using strainer or a piece of cloth. Keep it in a plastic container. BMO/IMO is ready to use after fermentation.

**Lactic Acid Bacteria Serum (LABS)**
After Fermenting for 1 week. Extract the liquid using strainer or a piece of cloth Keep it in a plastic container. BMO/IMO is ready to use after fermentation.

**Concoctions Dosage**

Fermentation

Application

**FPJ**

2tbsp./Lof water

Spray or drench 2x a week

7 days

**FFJ**

2 tbsp./L of water

Spray or drench 2x a week

7 days

**FAA**

2 tbsp./L of water

Spray or drench 2x a week

7 days

**OHN**

2 tbsp./L of water

Spray or drench 2x a week

13 days

**CALPHOS**

2 tbsp./L of water

Spray or drench 2x a week

37 days

**BMO/IMO**

2 tbsp./L of water

Spray or drench 2x a week

12 days

**LABS**

2 tbsp./L of water

Spray or drench 2x a week

28 days

## SELF CHECK 4.2-4
## Period of harvest

**Multiple Choice:** Choose the letter of your answer and write them on the space provided before the number.

_____ 1. The following are important task in producing organic fertilizers except one, which of them?
A. spraying synthetic chemicals
B. harvesting of fertilizer
C. preparing composting area and raw materials
D. carrying-out composting activities

_____ 2. The following statement describes the importance of organic concoctions?
A. safe for carcinogenic substance;
B. an environment-friendly;
C. many business opportunities
D. all of the above

_____ 3. Which of the following measures the correct dosage in using fermented concoction?
A. 1 tbsp./L of water
B. b. 2 tbsp./L of water
C. 10 tbsp. /L of water
D. 20 tbsp. /L of water

_____ 4. It is a nitrogen fertilizer and growth enhancer for plants and animals.
a. FAA
c. FFJ
b. FPJ
d. LABS

_____ 5. It is a potassium fertilizer for plants and taste enhancer to animals.
a. BMO/ IMO
c. FAA
b. CALPHOS
d. FFJ

# ANSWER KEY 4.2-4

## Period of harvest

**Multiple Choice**

1. A
2. D
3. B
4. B
5. D

# JOB SHEET 4.2-4

## Period of harvest

| |
|---|
| **Title: Determine the harvesting period of concoctions** |
| **Performance Objective:** At the end of this module you should be able to determine the harvesting period of concoction. |
| **Supplies/Materials:** PPE, various concoction, 1 liter plastic bottle |
| **Steps/Procedures:**<br><br>Wear the PPE<br>Harvest the concoction<br>Put in a sanitized and cleaned plastic bottle<br>Correct the measure dosage of using concoction<br>Do housekeeping. |
| **Assessment method:**<br>**Demonstration:** |

## PERFORMANCE CHECKLIST 4.2-4

## Period of harvest

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee... | | |
| Harvest the concoction? | | |
| Put inside in a sanitized and cleaned plastic bottle? | | |
| Correct the measure dosage of using concoction? | | |
| Do housekeeping? | | |

## Learning Outcome # 3 Package concoctions

**Assessment Criteria:**

Concoctions are contained in sanitized bottles and containers.

Packaged concoctions are labeled and tagged in accordance with enterprise practice.

Packaged concoctions are stored in appropriate place and temperature following organic practices.

Production of concoctions are recorded using enterprise procedures.

**Contents:**

Sanitize bottles and containers

Package concoctions

Appropriate place to store

Production Record

**Tools materials and equipment**:

Various concoctions

Concoction area

sanitized bottles

Labelling

Storing area

## LEARNING EXPERIENCES/ACTIVITIES

## LO # 3 Package concoctions

| Learning Activities | Special Instructions |
|---|---|
| Read the attached Information Sheet 4.3-1 Sanitize bottles and containers | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 4.3-1 on Sanitize bottles and containers | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 4.3-1 on Sanitize bottles and containers | |
| Perform Job Sheet 4.3-1 on Sanitize bottles and containers | Observe the demonstration of your trainer and check the procedures while performing the operation. |
| Evaluate performance using performance checklist on 4.3-1 on Sanitize bottles and containers | |
| Read the attached Information Sheet 4.3-2 on Package concoctions | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 4.3-2 on Package concoctions | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 4.3-2 on Package concoctions | |
| Perform Job Sheet 4.2-2 on Package concoctions | Observe the demonstration of your trainer and check the procedures while performing the operation. |

| | |
|---|---|
| Evaluate performance using performance checklist on 4.3-2 on Package of concoction | |
| Read the attached Information Sheet 4.3-3 on Appropriate place to store | You can ask the assistance of your trainer explain further the topics you cannot understand |
| Answer Self-Check 4.3-3 on Appropriate place to store | Try to answer the Self-Check without looking at the information sheet |
| Compare your answer to Answer Key 4.3-3 on Appropriate place to store | |
| Perform Job Sheet 4.3-3 on Appropriate place to store | Observe the demonstration of your trainer and check the procedures while performing the operation. |
| Evaluate performance using performance checklist on 4.3-3 on Appropriate place to store | |

## INFORMATION SHEET 4.3-1
### Sanitize bottles and containers

**Learning Objective:** After reading this information sheet, you should be able to sanitize the bottles and containers for concoctions.

**STERILIZATION**

Staff at production and research laboratories use high-pressure steam inside autoclave to sterilize or remove all microorganisms from plastic containers. These containers must be rated safe for an autoclave as some plastics, such as HDPE and polyethylene, will melt in the course of a standard autoclave run. For those looking to sterilize plastic containers at home, a standard microwave oven will do the trick. Of course, only microwave-safe plastics ought to be sterilized in this manner. Although not appropriate for home sterilization, plastic container sterilizations also be accomplished via ethylene oxide 'gas' sterilization, per acetic acid, ionizing graduation, dry heat, hydrogen peroxide gas plasma systems, ozone, formal dehyde steam, gaseous chlorine dioxide and infrared radiation.

**Microwave Sterilization**

**1.Prepare a Heat Sink**
Fill a cup with 250 to 500 ml (about 1 to 2 cups) of water and place it in the microwave. This will act as a heat sink to ensure the plastic container inside the microwave doesn't get too hot and melt.

**2. Place Containers in Microwave**
Gather together the microwave-safe containers and lids that require sterilization. Microwave containers in a secondary container for at least 3 minutes on the highest setting.

**3. Take out sterilized containers**
Remove secondary container for microwave with plastic containers inside, while maintaining sterility. Use insulated gloves, as the containers may be hot.

**Autoclave Sterilization**

**1. Prepare the Containers**
Gather together autoclave-safe plastic containers and any lids that need sterilization. Lids can be loosely placed on top of containers. A tightly attached

lid can cause a container to succumb to pressure within the autoclave and crack or explode.

**2. Organize the Containers**
Place containers and lids in a secondary autoclave-safe container, making sure to leave space between containers.

**3. Follow the Operating Procedures**
Place the secondary container in the autoclave and follow any standard operating procedures for your specific autoclave. The standard sterilizing autoclave run is at 121 degrees Celsius, 15 pounds per square inch of pressure for at least 30 minutes.

**4. Remove Sterilized Containers carefully**
Remove the secondary container from the autoclave using thick, insulated gloves to avoid burning. The surfaces will be extremely hot.

**How to sterilize plastic bottles**

**Step 1**
Remove all the labels from your bottles before cleaning them. You can do this with scissors or by soaking the bottle in warm soapy water to dilute the glue. The method you use will depend on the actual labels. For example, soft drink bottles have labels that are only glued in one spot. Using the scissors, you can easily cut the band and then unwrap what's left.

**Step 2**
Unscrew the tops and set them in a container of warm soapy water to avoid losing them down the drain.

**Step 3**
Fill a large pot or sink with soap and hot water. You'll want to fully submerge your bottles into the solution for a few minutes to kill any bacteria. If scratches are on the inside of the bottles, recycle them. Bacteria can become lodged in those scratches and multiply later.

**Step 4**
Rinse the bottles and tops thoroughly. For the bottles, fill them with warm water from the tap until no soap residue is left over. You can also place the caps on the bottles and shake the water inside to see if soap bubbles appear. If they do, rinse again.

**Step 5**
Stand the bottles upside down on a well-ventilated drying rack. Don't lay them on their sides as they'll take longer to dry and may dry unevenly. Allow the bottles to dry overnight. Don’t refill the bottles too soon; they must be completely

dry before refilling to avoid bacterial buildup.

**How to sterilize plastic bottles**

**Purpose To perform proper sterilization of plastic bottles**

**Supplies/Materials Plastic bottles, water, detergents**

**Procedure:**
1. Remove all the labels from your bottles before cleaning them. You can do this with scissors or by soaking the bottle in warm soapy water to dilute the glue.

2. Unscrew the tops and set them in a container of warm soapy water to avoid losing them down the drain.

3. Fill a large pot or sink with soap and hot water. You'll want to fully submerge your bottles into the solution for a few minutes to kill any bacteria. If scratches are on the inside of the bottles, recycle them.

4. Rinse the bottles and tops thoroughly. For the bottles, fill them with warm water from the tap until no soap residue is left over. You can also place the caps on the bottles and shake the water inside to see if soap bubbles appear. If they do, rinse again.

5. Stand the bottles upside down on a well-ventilated drying rack. Don't lay them on their sides as they'll take longer to dry and may dry unevenly. Allow the bottles to dry overnight. Don't refill the bottles too soon; they must be completely dry before refilling to avoid bacterial buildup.

# SELF CHECK 4.3-1

## Sanitize bottles and containers

**True or False**

1. Remove all the labels from your bottles before cleaning.
2. Unscrew the tops and set them in a container of warm soapy water to avoid losing them down the drain.
3. You will want to fully submerge your bottles into the solution for a few minutes to kill any bacteria.
4. You can also place the caps on the bottles and shake the water inside to see if soap bubbles appear.
5. Stand the bottles upside down on a well-ventilated drying rack. Don't lay them on their sides as they'll take longer to dry and may dry unevenly. Allow the bottles to dry overnight. Don't refill the bottles too soon

## ANSWER KEY 4.3-1
## Sanitize bottles and containers

### True or False

1. True
2. True
3. True
4. True
5. True

# JOB SHEET 4.3-1

## Sanitize bottles and containers

| |
|---|
| **Title: Sanitizing the bottles and containers** |
| **Performance Objective:** At the end of this module the student should be able to sanitize bottles and containers. |
| **Supplies/Materials: PPE, Concoction area, sanitized bottles and plastic bottle.** |
| **Steps/Procedures:**<br><br>Wear PPE<br>Remove the label before cleaning the plastic bottle<br>Unscrew the tops and of a container of warm soapy water to avoid losing them down the drain<br>Fill a large pot or sink with soap and hot water to fully submerge your bottles into the solution for a few minutes to kill any bacteria<br>Rinse the bottles and tops thoroughly. For the bottles, fill them with warm water from the tap until no soap residue is left over<br>Allow the bottles to dry overnight. Don't refill the bottles too soon; they must be completely dry before refilling to avoid bacterial buildup.<br>Do housekeeping |
| **Assessment Method:**<br><br>Demonstration |

# PERFORMANCE CHECKLIST 4.3-1

## Sanitize bottles and containers

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee… | | |
| 1. Wear PPE? | | |
| 2. Remove the label before cleaning the plastic bottle? | | |
| 3 Unscrew the tops and of a container of warm soapy water to avoid losing down the drain? | | |
| 4. Fill a large pot or sink with soap and hot water to fully submerge your bottles into the solution for a few minutes to kill any bacteria? | | |
| 5 Rinse the bottles and tops thoroughly. For the bottles, fill them with warm water from the tap until no soap residue is left over? | | |
| 6. Allow the bottles to dry overnight. Don't refill the bottles too soon; they must be completely dry before refilling to avoid bacterial buildup. | | |
| 7. Do housekeeping? | | |

## INFORMATION SHEET 4.3-2
## Package concoctions

**Learning Objective:** after reading this information sheet, you should be able to appreciate the proper labeling and packaging of concoctions

**Introduction:**

**PROPER LABELING AND PACKAGING**

Importance of labeling and packaging

another main purpose of the use of labeling and packaging is to exaggerate the product.

A marketer needs to grab the attention of a viewer to purchase the product. Labeling and packaging should be able to beautify a product to add to its visual appeal.
This can instantly grab a viewer's attention towards a product. You can arouse interest in the mind of a customer towards a product through an attractively designed label. It is essential to use a good quality material for the sticker.

A label needs to comply with the Competition and Consumer Act 2010 (CCA). This Act is required to give information to consumers, such as:
• The mandatory consumer product information standards under the CCA
• Industry specific regulations, such as the Food Standards Code
• Labels required by customs for some imported products under the Commerce (Trade

**Descriptions) Act.**
• The role of packaging and labeling has become quite significant as it helps to grab.

**Attention of the audience.**
• Labelling and packaging can be used by marketers to encourage potential buyers.

**Purchase the product.**
• Packaging is also used for convenience and information transmission.

Packages and labels communicate how to use, transport, recycle or dispose of the package or product.

**Name of**

- True nature of the Product
- state type of treatment it has undergone
- the name of the product shall be presented in bold type letters

- a complete list of raw materials shall Ingredients be declared in descending order of proportion

- A specific name, not collective (generic) name shall use as an ingredient

**The name**

- the name and address of the and address manufacturer, packer or the distributor of the of the food shall be declared in the manufacturer, label. Packer or distributor

- if the product is not manufactured by person or company whose appear in the label, the name must be qualified by manufactured for or packed for or similar expression.

**Manufacturing**

- the date when the product was and manufactured and the date of expiration shall be placed visibly within Date the label

## SELF CHECK 4.3-2

## Package concoctions

**True or False**

1. Labelling and packaging can be used by marketers to encourage potential buyers.

2. Packages and labels communicate how to use, transport, recycle or dispose of the package or product

3. The name and address of manufacturer, packer or the distributor of the of the food shall be declared in the manufacturer, label. Packer or distributor.

4. The product is not manufactured by person or company whose appear in the label, the name must be qualified by manufactured for or packed for or similar expression.

5. The date when the product was and manufactured and the date of expiration shall be placed visibly within Date the label

# ANSWER KEY 4.3-2

## Package concoctions

**True or False**

1. True
2. True
3. True
4. True
5. True

# JOB SHEET 4.3-2

## Package concoctions

| |
|---|
| **Title:** Proper labeling and packaging of concoctions |
| **Performance Objective:** At the end of this module the student should be able to proper labeling and packaging of concoctions. |
| **Supplies/Materials:** **concoction area, various concoction, papers, ballpeen.** |
| **Steps/Procedures:**<br><br>Prepare all the supplies and materials in labelling and tagging<br>Get your produce concoctions<br>Label the fermented products based on the required information<br>Submit your work to your trainer.<br>Do housekeeping |
| **Assessment Method:**<br><br>Demonstration |

## PERFORMANCE CHECKLIST 4.3-2

## Package concoctions

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee… | | |
| Prepare all the supplies and materials in labelling and tagging? | | |
| Get your produce concoctions? | | |
| Label the fermented products based on the required information? | | |
| Submit your work to your trainer? | | |
| Do housekeeping? | | |

## INFORMATION SHEET 4.3-3
## Appropriate place to store

**Learning Objective:** After reading this information sheet, you should be able to determine the appropriate storage for the various concoctions.

**Introduction:**

Storage Requirements of various concoctions

Store the covered container in a well-ventilated area away from artificial or natural light and extreme heat or cold. Do not refrigerate. In order for the fermentation process to occur properly, the volume of the plant-material-and-brown-sugar mixture should settle to 2/3 of the container after 24 hours.

**Storage Requirements of various concoctions**

**Fermented Fruit Juice**

Store the container with the bagged mixture for 7 days in a cool dry shady place. Make sure that the storage area is not infested with cockroaches or mice, because they might feed on the mixture and contaminate the extracts. In 7 days, plant juice is extracted and fermented. The fruit extract will change its color from yellow orange to brown and will smell sweet and alcoholic. After 7 days, lift the bagged mixture and squeeze hard to get the remaining extracts.

**Fermented Plant Juice**

Polyethylene or glass products or clay jar may be used as a container. When using glass bottles, brown glass containers must be preferred.

Store in a cool place. Select a shaded area where there is no direct sunlight and where the temperature does not fluctuate. Direct sunlight should be avoided.

The optimum temperature range is 1 to15°C for storage (Use a Refrigerator if available) if you want to keep for one year. Otherwise one can use within 30 days store at room temperature.

**Fish Amino Acid**

Store the container with the mixture for 4 weeks in a cool dry shady place. Make sure that the storage area is not infested with cockroaches or mice because they might feed on the mixture and contaminate the extract. The mixture may be appealing to the house pets so make sure that it is properly secured. In a month's time, the fermented extract is ready.

**Lactic Acid Bacteria Serum (LABS)**

Keep the refined LAB serum at cool temperature, so for longer period where there is temperature change (1-15°C ). No storage under direct sunlight.

In order to keep LAB at a normal temperature it must be mixed with the same amount of brown sugar and stirred with a wooden stick (ladle).
Note: Using rice-washed water in obtaining lactic acid bacteria is to collect stronger ones. Only strong ones can survive in poor nutrients condition like rice washed water.

**Oriental Herbs Nutrients (OHN)**
The jar must cover it with tight lid / vinyl film. Stir the mixture gently clockwise every day morning for a week. Leave it for 4-6 days. Filter the content and keep the extraction in another jar for long-term storage.

The extracting process is difficult add water to extract juice this can be used within 45 days.

**Indigenous Microorganism**

Keep the IMO3 bags in shaded and cool place. Make sure that the air is well circulated by keeping IMO-3 in a ventilated container such as jute / gunny / cloth bags.

First, spread rice straw or leaf litter at the bottom of the container, and put in IMO-3. During storage, the IMO-3 may become dry (moisture level 20-30%) as the moisture gets evaporated. It means that the IMOs have entered a sleeping phase (state of dormancy). Pile up containers into 3 layers and shield them from direct sunlight and rain. At this point, there is no need to turn over, because of the convection currents that are created through the gaps of containers.

## TEMPERATURE AND SHELFLIFE REQUIREMENT

### Temperature Control

There should be active mechanical temperature control and no direct sources of heat (sunny windows, steam pipes, furnaces, etc.).

Most are fairly stable at room temperature. 4 degrees Celsius should be ok for most plant extracts. I would also suggest keeping it out of light as well. As light can alter certain compound and decrease their potency and change their structure, Shelf life.

**Various concoctions and extracts packaged in bottles/ containers**

- The sealed bottles or other package prevents contamination during storage and transport.
- The concoctions/extracts will have a long but not an indefinite storage life.
- The storage life depends on several factors, but especially on the nature of the product and the conditions of storage.
- Concoctions have a storage life for many years.
- As a general rule, the lower the storage temperature, the longer the storage life will be.

## SELF CHECK 4.3-3

## Appropriate place to store

### True or Falls

**1**. Fish Amino Acid- Store the container with the mixture for 4 weeks in a cool dry shady place. Make sure that the storage area is not infested with cockroaches or mice because they might feed on the mixture and contaminate the extract.

2. Lactic Acid Bacteria Serum (LABS) - Keep the refined LAB serum at cool temperature, so for longer period where there is temperature change (1-15°C). No storage under direct sunlight.

3. Indigenous Microorganism- Keep the IMO3 bags in shaded and cool place. Make sure that the air is well circulated by keeping IMO-3 in a ventilated container such as jute / gunny / cloth bags

4. Oriental Herbs Nutrients (OHN)- The jar must cover it with tight lid / vinyl film. Stir the mixture gently clockwise every day morning for a week. Leave it for 4-6 days. Filter the content and keep the extraction in another jar for long-term storage.

5. Fermented Fruit Juice- Store the container with the bagged mixture for 7 days in a cool dry shady place. Make sure that the storage area is not infested with cockroaches or mice, because they might feed on the mixture and contaminate the extracts.

# ANSWER KEY 4.3-3

## Appropriate place to store

### True or False

1. True
2. True
3. True
4. True
5. True

## JOB SHEET 4.3-3

## Appropriate place to store

| |
|---|
| **Title: Appropriate place to store** |
| **Performance Objective:** At the end of this module the student should be able to appropriate storage for the various concoctions. |
| **Supplies/Materials:** Concoction area, various concoction |
| **Steps/Procedures:**<br><br>Prepare the various concoctions<br>Determine the right temperature inside the concoction area<br>Sealed bottles or other package prevents contamination during storage<br>Do housekeeping |
| **Assessment Method:**<br><br>Demonstration |

# PERFORMANCE CHECKLIST 4.3-3

## Appropriate place to store

| Performance Criteria | YES | NO |
|---|---|---|
| Did the trainee… | | |
| Prepare the various concoctions? | | |
| Determine the right temperature inside the concoction area? | | |
| Sealed bottles or other package prevents contamination during storage? | | |
| Do housekeeping? | | |